Exercise Sheet 10 (Typed in LaTeX)

English subpart labels: (a), (b), (c), ...

1. Compute the following integrals

(a) $\int \left(2x + 3e^x - x\sqrt{x}\right)\, dx$

(b) $\int \left(5\cos x + 3^x + \dfrac{1}{2\sqrt[3]{x}}\right)\, dx$

(c) $\int \dfrac{(18+2x^2)\sin x + 5}{x^2+9}\, dx$

(d) $\int \dfrac{(\sqrt{x}+2)^2}{\sqrt{x}}\, dx$

(e) $\int \dfrac{3-\cos^2 x}{\cos^2 x}\, dx$

(f) $\int \sqrt[3]{5-6x}\, dx$

(g) $\int \dfrac{2x^3+8x-5}{4+x^2}\, dx$

(h) $\int 3^{\,2-3x}\, dx$

(i) $\int \left(\dfrac{9}{9+x^2} + \dfrac{5}{x^2}\right)\, dx$

(j) $\int \dfrac{dx}{1+2x^2}$

(k) $\int \dfrac{dx}{\sqrt{36-3x^2}}$

2. Compute the following integrals

(a) $\int e^{-2x+1}\, dx$

(b) $\int \dfrac{dx}{1-3x}$

(c) $\int \dfrac{\tan^2 x + 3x\sin^2 x}{\sin^2 x}\, dx$

(d) $\int \dfrac{dx}{9-4x^2}$

(e) $\int \dfrac{1+x+3x^2}{\sqrt{x}}\, dx$

(f) $\int \left(e^{5x} - \dfrac{2x^2+5}{1+x^2}\right)\, dx$

(g) $\int \dfrac{dx}{(5-3x)^2}$

(h) $\displaystyle\int \frac{dx}{3-16x^2}$

(i) $\displaystyle\int \frac{dx}{\sqrt{4-3x^2}}$

(j) $\displaystyle\int \frac{dx}{\sqrt{1-(4x+2)^2}}$

(k) $\displaystyle\int \frac{dx}{1-5x^2}$

3. Compute the following integrals (use substitution)

(a) $\displaystyle\int \frac{4x^3-1}{x^4-x+5}\,dx$

(b) $\displaystyle\int \frac{\sqrt[4]{\ln x}}{x}\,dx$

(c) $\displaystyle\int \frac{3x}{(x^2+4)^3}\,dx$

(d) $\displaystyle\int \frac{e^{\tan x}}{\cos^2 x}\,dx$

(e) $\displaystyle\int \frac{\cos x}{\sqrt{\sin^2 x+3}}\,dx$

(f) $\displaystyle\int \frac{\sqrt{\arctan x}}{1+x^2}\,dx$

(g) $\displaystyle\int \sqrt[3]{2+7\sin x}\;\cos x\,dx$

(h) $\displaystyle\int \frac{1}{x^2}\cos\left(\frac{1}{x}\right)\,dx$

(i) $\displaystyle\int \frac{\cos(5x)}{\sin^2(5x)+1}\,dx$

(j) $\displaystyle\int \frac{\ln x+2}{2x}\,dx$

(k) $\displaystyle\int \frac{dx}{\sin^2 x\,\sqrt{3-\cot x}}$

4. Compute the following integrals (integration by parts)

(a) $\displaystyle\int e^{-x}(x^2+5)\,dx$

(b) $\displaystyle\int \arcsin x\,dx$

(c) $\displaystyle\int x\sin(3x)\,dx$

(d) $\displaystyle\int e^{x}\sin(3x)\,dx$

(e) $\int x \cdot 10^x \, dx$

(f) $\int (x-2)\ln(2x)\, dx$

(g) $\int \arccos(2x)\, dx$

(h) $\int \ln(x+2)\, dx$

(i) $\int \cos(\ln x)\, dx$

(j) $\int x^2 \ln x \, dx$

(k) $\int x^2 \cos^2 x \, dx$

5. Compute the following partial-fractions-type integrals

(a) $\int \frac{10\, dx}{15+4x}$

(b) $\int \frac{dx}{4-7x}$

(c) $\int \frac{dx}{(x+15)^{20}}$

(d) $\int \frac{dx}{(5-2x)^8}$

(e) $\int \frac{x}{2x^2+2x+5}\, dx$

(f) $\int \frac{x}{x^2-7x+20}\, dx$

(g) $\int \frac{x+1}{5x^2+2x+1}\, dx$

(h) $\int \frac{2x+1}{x^2-4x+5}\, dx$

(i) $\int \frac{dx}{2x^2-2x+3}$

6. Compute the following rational-function integrals (use partial fractions)

(a) $\int \frac{3x^3+5x^2-25x-1}{(x+2)(x-1)^2}\, dx$

(b) $\int \frac{x^2-3x+17}{(x+5)(x^2+2x+4)}\, dx$

(c) $\int \frac{x^2+3}{(x+1)^2(x-1)}\, dx$

(d) $\int \frac{x^3+2x+5}{x^2-1}\,dx$

(e) $\int \frac{x^3+3x^2+5x+7}{x^2+2}\,dx$

(f) $\int \frac{x^2+2x+6}{(x-1)(x-2)(x-4)}\,dx$

(g) $\int \frac{2x^2+x+3}{(x+2)(x^2+x+1)}\,dx$

(h) $\int \frac{dx}{x^5-x^2}$

7. Compute the following trigonometric integrals

(a) $\int \frac{\cos x}{1+\cos x}\,dx$

(b) $\int \frac{dx}{4\sin x+3\cos x+5}$

(c) $\int \frac{dx}{5\sin x+3\cos x+3}$

(d) $\int \sin^5 x\,dx$

(e) $\int \sqrt[3]{\sin^2 x}\,\cos^3 x\,dx$

(f) $\int \sin^2 x\,\cos^2 x\,dx$

(g) $\int \sin^4(2x)\,dx$

(h) $\int \sin(5x)\cos x\,dx$

(i) $\int \cos(3x)\cos(2x)\,dx$

8. Compute the following integrals (irrational expressions)

(a) $\int \frac{x+1}{\sqrt[3]{3x+1}}\,dx$

(b) $\int \frac{dx}{\sqrt[3]{(2x+1)^2}-\sqrt{2x+1}}$

(c) $\int \frac{\sqrt[6]{x}}{1+\sqrt[3]{x}}\,dx$

(d) $\int \frac{7x+16}{\sqrt[3]{(7x+8)^2}+2\sqrt[3]{7x+8}}\,dx$